Lab 2 – KKT Conditions and Constrained Optimization

In this lab you will use **Karush–Kuhn–Tucker (KKT)** conditions to analyze and solve constrained optimization problems. You will combine analytical derivations with numerical and graphical checks in MATLAB (or Python).

All algebraic and analytical steps must be performed programmatically using MATLAB's symbolic toolbox or the corresponding Python libraries; **no hand-calculations are required or expected**.

Objectives

- Formulate constrained optimization problems in standard form.
- Set up and solve KKT conditions for equality and inequality constraints.
- Determine active constraints and interpret Lagrange multipliers.
- Understand when KKT conditions are necessary but not sufficient.
- Verify solutions graphically in low-dimensional cases.

What to do for each problem

For each problem listed below, use MATLAB or Python to:

- Write the objective function and constraints in standard form (inequalities as `g(x) ≤ 0`, equalities as `h(x) = 0`).
- Formulate the Lagrangian.
- Derive all KKT conditions.
- Determine possible active sets, solve for candidate KKT points, and check feasibility.
- Classify each feasible KKT point (minimum / maximum / saddle) using second-order conditions or graphical reasoning when appropriate.
- Compute Lagrange multipliers and, for inequality constraints, the corresponding slack and squared slack variables.
- Create suitable plots (e.g. level curves and constraint lines/regions) for the two-dimensional problems and mark the candidate solutions.

Recommended reading and tools

Arora, Introduction to Optimum Design, 3rd ed.

- Chapter 4, Sections 4.5.1–4.5.2 (KKT conditions)
- Chapter 4, Section 4.6 (Constraint activity)
- Chapter 4, Section 4.7 (Lagrange multipliers)
- Chapter 4, Section 4.8 (Second-order conditions and examples)

Helpful MATLAB functions

- syms, diff, solve
- gradient, hessian
- meshgrid, contour, plot

Optional Python tools (NumPy / SymPy / Matplotlib)

- sympy.symbols, sympy.diff, sympy.solve, sympy.hessian
- numpy.meshgrid
- matplotlib.pyplot.contour, matplotlib.pyplot.plot

Problems

1. Maximize
   $F(x_1, x_2) = 4x_1^2 + 3x_2^2 - 5x_1x_2 - 8$
   subject to
   $x_1 + x_2 \leq 4$.

2. Maximize
   $F(x_1, x_2) = 4x_1^2 + 3x_2^2 - 5x_1x_2 - 8x_1$
   subject to
   $x_1 + x_2 \leq 4$.

3. Minimize
   $f(x_1, x_2) = (x_1 - 1)^2 + (x_2 - 1)^2$
   subject to
   $x_1 + x_2 \geq 4$,
   $x_1 - x_2 - 2 = 0$.

4. Minimize
   $f(x_1, x_2) = 4x_1^2 + 3x_2^2 - 5x_1x_2 - 8x_1$
   subject to
   $x_1 + x_2 = 4$.

5. Minimize
   $f(x_1, x_2, x_3) = (x_1 - 2)^2 + (x_2 - 1)^2 + (x_3 - 3)^2$
   subject to
   $x_1 + x_2 + x_3 \geq 6$,
   $x_1 - 2x_2 \leq 1$,
   $x_3 \geq 1$.

   Pay particular attention to possible combinations of active constraints and how they affect the KKT system.

6. Minimize
   $f(x) = x^4 - 2x^2$
   (unconstrained problem).

   In addition to the general steps, explicitly identify all stationary points and show which of them correspond to local minima, local maxima, or saddle-type behavior, to illustrate that satisfying the KKT (stationarity) condition is not sufficient for optimality in a nonconvex problem.

7. Minimize
   $f(x_1, x_2) = (x_1 - 3)^2 + (x_2 - 2)^2$
   subject to
   $x_1 \geq 2$,
   $x_2 \geq 1$,
   $x_1 + x_2 \leq 4$.

   For this problem, report for the final solution:

   - Which constraints are active and which are inactive.
   - The corresponding Lagrange multipliers.
   - Slack and squared slack variables for each inequality constraint.